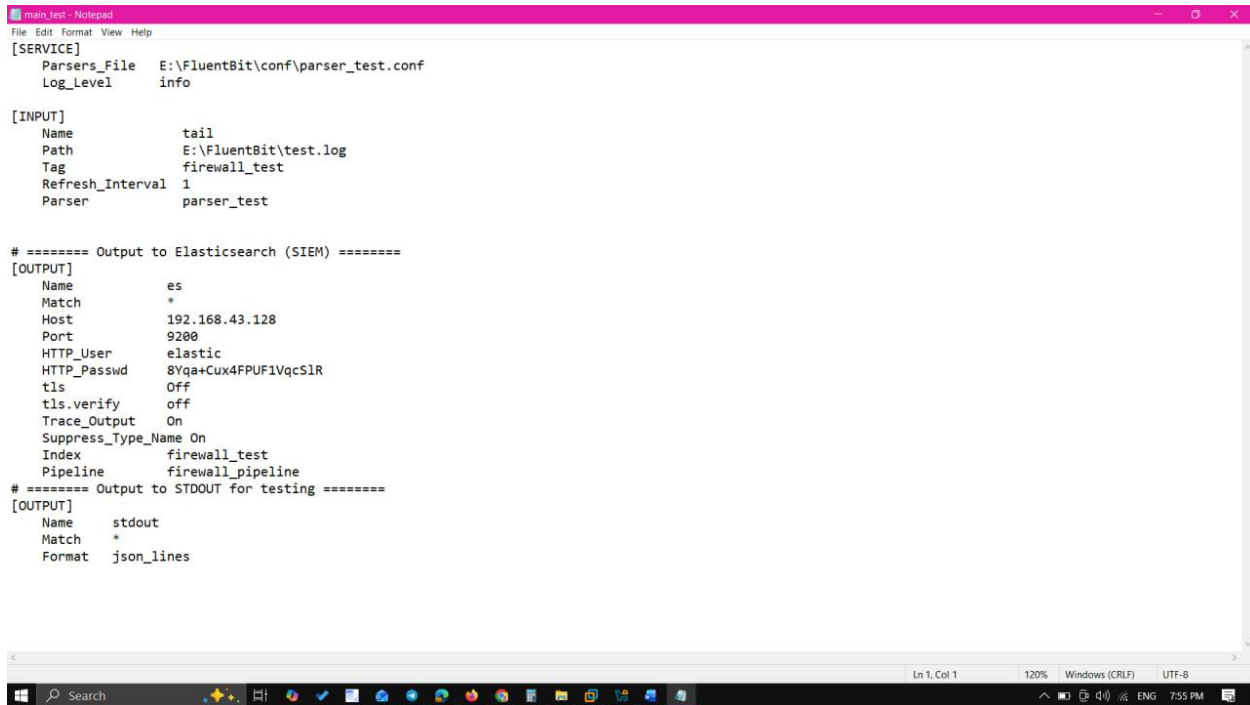


Install and configure fluentbit

1. Configuring Fluent Bit Input and Output

Fluent Bit is configured using a configuration file that defines data sources (inputs) and destinations (outputs). The screenshot shows both an input and output configuration.



```
[SERVICE]
  Parsers_File  E:\FluentBit\conf\parser_test.conf
  Log_Level     info

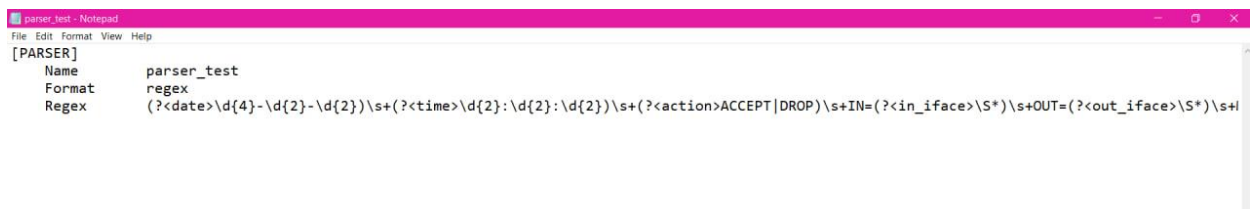
[INPUT]
  Name          tail
  Path          E:\FluentBit\test.log
  Tag           firewall_test
  Refresh_Interval 1
  Parser        parser_test

# ===== Output to Elasticsearch (SIEM) =====
[OUTPUT]
  Name          es
  Match         *
  Host          192.168.43.128
  Port          9200
  HTTP_User     elastic
  HTTP_Passwd   8Yqa+Cux4FPuF1VqcS1R
  tls           Off
  tls.verify    off
  Trace_Output  On
  Suppress_Type_Name On
  Index         firewall_test
  Pipeline      firewall_pipeline

# ===== Output to STDOUT for testing =====
[OUTPUT]
  Name          stdout
  Match         *
  Format        json_lines
```

2. Configuring Fluent Bit Parser with Regular Expressions

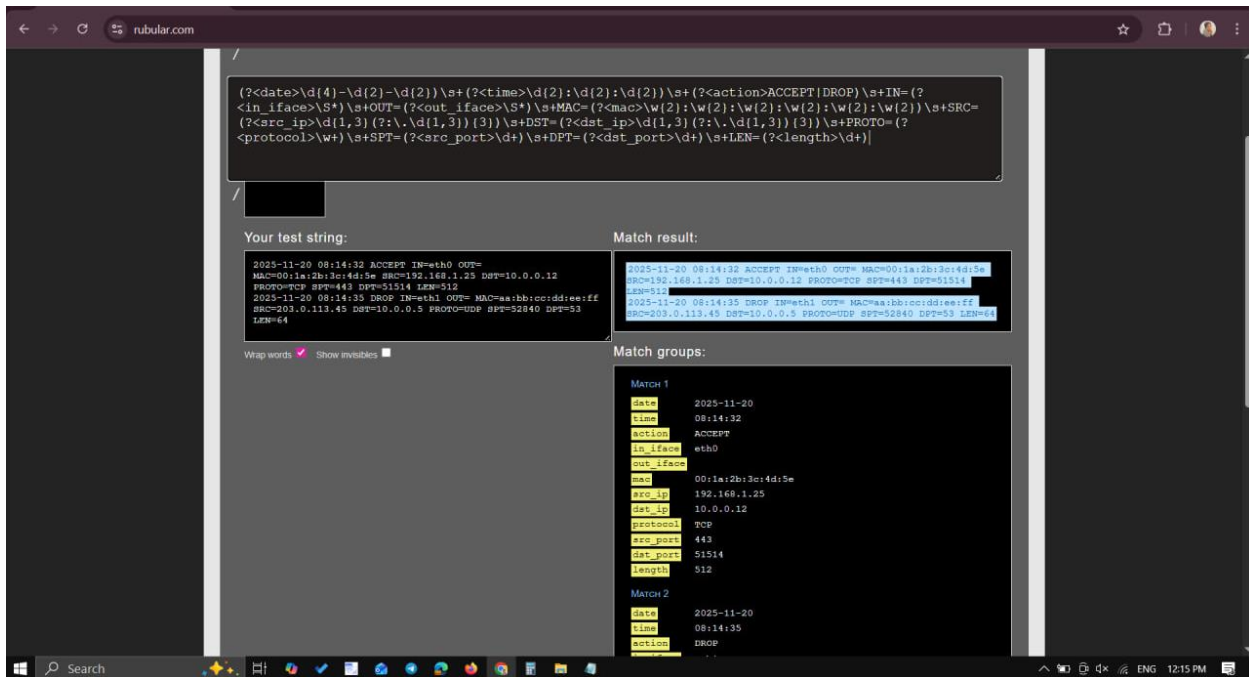
Parsers in Fluent Bit extract structured fields from unstructured log lines. The screenshot shows a regex-based parser for firewall logs.



```
[PARSER]
  Name          parser_test
  Format        regex
  Regex         (?<date>\d{4}-\d{2}-\d{2})\s+(?<time>\d{2}:\d{2}:\d{2})\s+(?<action>ACCEPT|DROP)\s+IN=(?<in_iface>\S*)\s+OUT=(?<out_iface>\S*)\s+
```

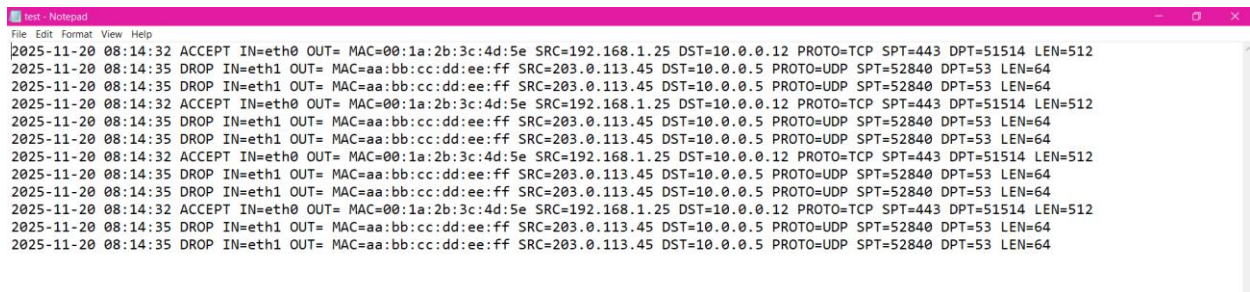
3. Testing Fluent Bit Regex Parser with Firewall Log Samples

Before deploying a parser in production, it is essential to test the regular expression against sample log lines to ensure correct field extraction. The screenshot shows a regex parser being tested with firewall log samples.



4.Raw Firewall Log Samples for Testing and Parser Validation

The screenshot shows two sample firewall log lines in a standardized format. These logs represent network traffic events that have been allowed (ACCEPT) or blocked (DROP) by a firewall. These samples are ideal for testing Fluent Bit parsers before deployment.



5.Starting Fluent Bit and Verifying Service Initialization

Fluent Bit is started by executing the binary with a configuration file. The command and startup logs are shown in the screenshot.

Command used:

```
.\fluent-bit.exe -c "E:\FluentBitHub\main_test.conf"
```

```
Administrator Windows PowerShell
PS C:\WINDOWS\system32> cd E:\FluentBit\bin
PS E:\FluentBit\bin> .\fluent-bit.exe -c E:\FluentBit\conf\main_test.conf
Fluent Bit v4.2.0
* Copyright (C) 2015-2025 The Fluent Bit Authors
* Fluent Bit is a CNCF graduated project under the Fluent organization
* https://fluentbit.io

Fluent Bit v4.2.0 [O Direct Routes Ahead
Celebrating 10 Years of Open, Fluent Innovation!

[2025/11/23 19:02:07.558362600] [info] [fluent bit] version=4.2.0, commit=6bc014390c, pid=23396
[2025/11/23 19:02:07.559396000] [info] [storage] ver=1.5.4, type=memory, sync=normal, checksum=off, max_chunks_up=128
[2025/11/23 19:02:07.559446700] [info] [simd] disabled
[2025/11/23 19:02:07.559474300] [info] [metrics] version=1.0.5
[2025/11/23 19:02:07.559501600] [info] [ctraces] version=0.6.6
[2025/11/23 19:02:07.560447200] [info] [input:tail:tail.0] initializing
[2025/11/23 19:02:07.560574300] [info] [input:tail:tail.0] storage_strategy='memory' (memory only)
[2025/11/23 19:02:07.560747900] [info] [output:es:es.0] worker #0 started
[2025/11/23 19:02:07.569161600] [info] [output:es:es.0] worker #1 started
[2025/11/23 19:02:07.571073800] [info] [sp] stream processor started
[2025/11/23 19:02:07.571358800] [info] [output:stdout:stdout.1] worker #0 started
[2025/11/23 19:02:07.572753700] [info] [engine] Shutdown Grace Period=5, Shutdown Input Grace Period=2
{"date":"1763917390.084972","log":"\r"}
{"date":"1763917390.085026","log":"\r"}
{"create":{"_index":"firewall_test"}}
{"@timestamp":"2025-11-23T17:03:10.004Z","log":"\r"}
{"create":{"_index":"firewall_test"}}
{"@timestamp":"2025-11-23T17:03:10.005Z","log":"\r"}
{"date":"2025-11-20","time":"08:14:32","action":"ACCEPT","in_iface":"eth0","mac":"00:1a:2b:3c:4d:5e","src_ip":"192.168.1.25","dst_ip":"10.0.0.12","protocol":"TCP","src_port":"443","dst_port":"51514","length":"512"}
{"date":"2025-11-20","time":"08:14:35","action":"DROP","in_iface":"eth1","mac":"aa:bb:cc:dd:ee:ff","src_ip":"203.0.113.45","dst_ip":"10.0.0.5","protocol":"UDP","src_port":"52840","dst_port":"53","length":"64"}
{"date":"1763917455.07569","log":"\r"}
{"date":"1763917455.075702","log":"\r"}
{"date":"1763917455.075706","log":"\r"}
{"date":"1763917455.075709","log":"\r"}
{"date":"1763917455.075712","log":"\r"}
{"date":"1763917455.075716","log":"\r"}
{"date":"1763917455.07572","log":"\r"}
{"date":"1763917455.075723","log":"\r"}
{"date":"1763917455.075727","log":"\r"}
{"date":"1763917455.07573","log":"\r"}
{"date":"1763917455.075734","log":"\r"}
{"date":"1763917455.075737","log":"\r"}
{"create":{"_index":"firewall_test"}}
{"@timestamp":"2025-11-23T17:04:15.075Z","date":"2025-11-20","time":"08:14:32","action":"ACCEPT","in_iface":"eth0","mac":"00:1a:2b:3c:4d:5e","src_ip":"192.168.1.25","dst_ip":"10.0.0.12","protocol":"TCP","src_port":"443","dst_port":"51514","length":"512"}
{"create":{"_index":"firewall_test"}}
{"@timestamp":"2025-11-23T17:04:15.075Z","date":"2025-11-20","time":"08:14:35","action":"DROP","in_iface":"eth1","mac":"aa:bb:cc:dd:ee:ff","src_ip":"203.0.113.45","dst_ip":"10.0.0.5","protocol":"UDP","src_port":"52840","dst_port":"53","length":"64"}

```

6.Verifying Fluent Bit Data in Elasticsearch Index Management

After Fluent Bit starts successfully and processes log data, a new index appears in Elasticsearch. The screenshot shows the Elasticsearch Index Management interface in Kibana.

